

Markov Decision Processes and Dynamic Programming

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Lecture Notes

Markov Decision Processes and Dynamic Programming

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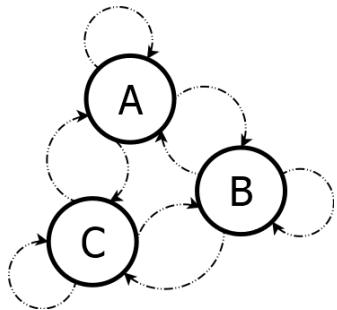
Markov Property

- Simply stated:
Given the current state, future observations are independent of past states, decisions, and outcomes.
- Future outcomes depend only on the current state, the decision taken, and subsequent states and actions.
- Also known as the *memoryless* property; it emphasizes focusing on the present and future while disregarding past events.
- Processes that satisfy this property are called Markov processes.
- Although primarily defined for stochastic systems, it can also be applied to deterministic settings.



Markov Chains

- In mathematics and optimization, it is defined as a *discrete-time stochastic process* that satisfies the Markov property:
 - “Discrete-time” means that state transitions occur at distinct time steps rather than continuously.
- In other words, it is a sequence of stochastic states where the probability of the next state depends only on the current state.

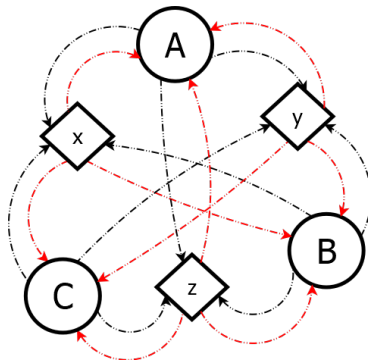


**Circles represent states,
arrows represent transitions.*



Markov Decision Processes

- In mathematics and optimization, it is defined as a *discrete-time stochastic control process*:
 - “Discrete-time” indicates that decisions are made sequentially at distinct time steps.
- It can be viewed as an extension of Markov chains in which decisions (actions) and rewards (outcomes) are incorporated.



**Circles represent states,
diamonds represent actions,
arrows represent transitions.*

*Red arrows (actions) also generate
outcomes.*



Markov Decision Processes

- In Operations Research, a Markov Decision Process (MDP) is a modeling framework for sequential decision-making.
- Systems with finite or infinite horizons, and with deterministic or stochastic, recurrent or transient transitions, can be modeled as MDPs.
- One of the most widely used solution approaches for MDPs is Dynamic Programming.



Application Areas

- Shortest path problems
- Inventory management problems
- Routing problems
- Cargo loading problems
- ...
- Any problem with a repetitive or recursive structure



Terminology

Stage A subpart of the overall problem

State A set of conditions describing the system at a given stage

Action A decision or move taken in a given state

Outcome The result of taking a specific action in a given state



Value Function

- The total expected value obtained from the current state onward when a particular policy is followed.



MDP Model

To construct an MDP model, the following must be defined:

- stages,
- states,
- actions,
- outcome (reward/cost) function,
- transition function (which determines the next state given the current state and action).



Principle of Optimality

- At any state, an optimal policy must maximize (or minimize) future outcomes based solely on the current state, independent of past states or decisions.



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Definition

- Dynamic Programming is a solution method that finds the optimal solution to a multistage decision problem by decomposing it into smaller subproblems, each involving a single decision variable.



Short History

- The method is commonly attributed to Bellman (1950).
- Earlier contributions by Hamilton and Jacobi also introduced similar recursive formulations.
- It is widely known as the Hamilton-Jacobi-Bellman (HJB) equation.
- Some studies suggest that similar ideas were used even earlier by Gauss.



Hamilton-Jacobi-Bellman Equation

$$V(s) = \min_a \{IC_s^a + V(s')\} \quad (1)$$

- s denotes the current state;
- $V(s)$ is the value function of state s ;
- a denotes an action;
- IC_s^a is the immediate cost of taking action a in state s ;
- s' denotes the next state resulting from action a .



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A More Detailed HJB Equation

$$V_t(s) \leftarrow \min_a \{IC_s^a + V_{t+1}(s')\} \quad (2)$$

- s denotes a state at stage t ;
- $V_t(s)$ is the value function of being in state s at stage t (i.e., the total outcomes from that stage until the end);
- a denotes a decision (action);
- IC_s^a represents the immediate outcome observed at stage t when action a is taken in state s ; and
- s' denotes the state encountered at stage $t + 1$ after taking action a in state s .

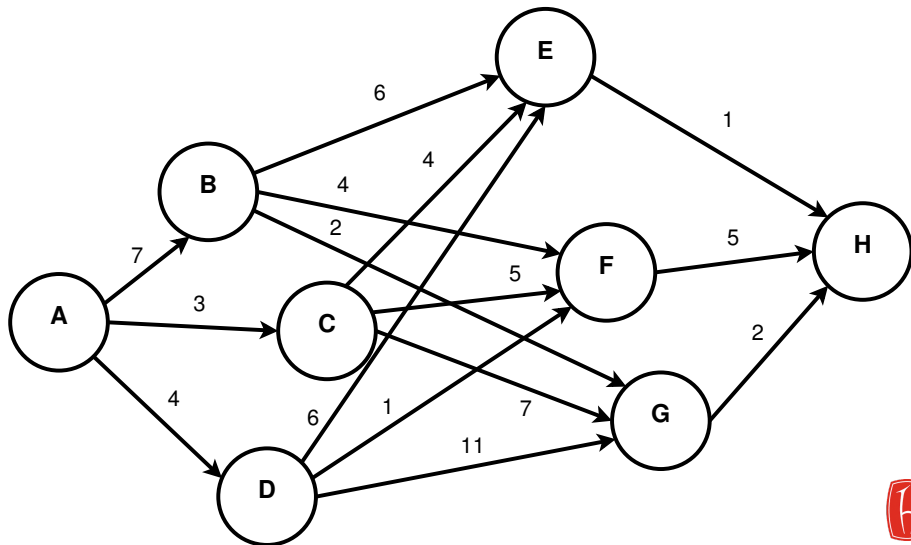


Backward Recursion

- The value function ($V_t(s)$) represents the total outcomes accumulated from stage t to the end of the *planning horizon*, given that the system is in state s at stage t .
- In other words: Value function = total cost/benefit from the current stage to the end.
- s' represents the state at stage $t + 1$ reached after taking action a in state s at stage t .



An Example



Backward Recursion Solution

$$V_4^*(H) = 0 + 0 = 0$$



Backward Recursion Solution

$$V_3^*(E) = IC_E^H + V_4^*(H) = 1 + 0 = 1$$

$$V_3^*(F) = IC_F^H + V_4^*(H) = 5 + 0 = 5$$

$$V_3^*(G) = IC_G^H + V_4^*(H) = 2 + 0 = 2$$



Backward Recursion Solution

$$V_2^*(B) = \min \left\{ \begin{array}{l} IC_B^E + V_3^*(E) = 6 + 1 = 7 \\ IC_B^F + V_3^*(F) = 4 + 5 = 9 \\ IC_B^G + V_3^*(G) = 2 + 2 = 4 \end{array} \right\} = 4$$



Backward Recursion Solution

$$V_2^*(C) = \min \left\{ \begin{array}{l} IC_C^E + V_3^*(E) = 4 + 1 = 5^* \\ IC_C^F + V_3^*(F) = 5 + 5 = 10 \\ IC_C^G + V_3^*(G) = 7 + 2 = 9 \end{array} \right\} = 5$$



Backward Recursion Solution

$$V_2^*(D) = \min \left\{ \begin{array}{l} IC_D^E + V_3^*(E) = 6 + 1 = 7 \\ IC_D^F + V_3^*(F) = 1 + 5 = 6^* \\ IC_D^G + V_3^*(G) = 11 + 2 = 13 \end{array} \right\} = 6$$



Backward Recursion Solution

$$V_1^*(A) = \min \left\{ \begin{array}{l} IC_A^B + V_2^*(B) = 7 + 4 = 11 \\ IC_A^C + V_2^*(C) = 3 + 5 = 8^* \\ IC_A^D + V_2^*(D) = 4 + 6 = 10 \end{array} \right\} = 8$$



Forward Recursion

$$V_t^*(s) \leftarrow \min_a \{IC_s^a + V_{t-1}^*(s')\} \quad (3)$$

- The value function ($V_t(s)$) represents the cumulative outcomes from the **beginning of the planning horizon** up to stage t , given that the system is in state s at stage t .
- In other words: Value function = cumulative cost/benefit from the beginning.
- s' represents the state at stage $t - 1$ from which state s is reached by taking action a .



Principle of Optimality

- At any state, an optimal policy must optimize past cumulative outcomes based solely on the characteristics of the current state, without considering how that state was reached or future decisions.



Forward Recursion Solution

$$V_1^*(A) = 0 + 0$$



Forward Recursion Solution

$$V_2^*(B) = IC_B^{AB} + V_1^*(A) = 7 + 0 = 7$$

$$V_2^*(C) = IC_C^{AC} + V_1^*(A) = 3 + 0 = 3$$

$$V_2^*(D) = IC_D^{AD} + V_1^*(A) = 4 + 0 = 4$$



Forward Recursion Solution

$$V_3^*(E) = \min \left\{ \begin{array}{l} IC_E^{BE} + V_2^*(B) = 6 + 7 = 13 \\ IC_E^{CE} + V_2^*(C) = 4 + 3 = 7^* \\ IC_E^{DE} + V_2^*(D) = 6 + 4 = 10 \end{array} \right\} = 7$$



Forward Recursion Solution

$$V_3^*(F) = \min \left\{ \begin{array}{l} IC_F^{BF} + V_2^*(B) = 4 + 7 = 11 \\ IC_F^{CF} + V_2^*(C) = 5 + 3 = 8 \\ IC_F^{DF} + V_2^*(D) = 1 + 4 = 5^* \end{array} \right\} = 5$$



Forward Recursion Solution

$$V_3^*(G) = \min \left\{ \begin{array}{l} IC_G^{BG} + V_2^*(B) = 2 + 7 = 9^* \\ IC_G^{CG} + V_2^*(C) = 7 + 3 = 10 \\ IC_G^{DG} + V_2^*(D) = 11 + 4 = 15 \end{array} \right\} = 9$$



Forward Recursion Solution

$$V_4^*(H) = \min \left\{ \begin{array}{l} IC_H^{EH} + V_3^*(E) = 1 + 7 = 8^* \\ IC_H^{FH} + V_3^*(F) = 5 + 5 = 10 \\ IC_H^{GH} + V_3^*(G) = 2 + 9 = 11 \end{array} \right\} = 8$$



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Cargo Loading Problems

- Cargo loading problems involve a storage/transportation unit with a specific capacity.
- Classically, there are specific items that can be loaded into this area.
- Decision makers aim to fill this space with the most valuable items.



Objective Function of the Problem

- The goal is to maximize the total value of the items placed in the area.
- The value of the items can be measured in monetary terms or in terms of the utility provided to the decision maker.



An Example

A 4-ton truck will be loaded with one or more of three products. The first product weighs 2 tons per unit, the second product weighs 3 tons, and the third product weighs 1 ton. Each unit of the first product yields a \$31 profit, the second product yields \$47, and the third product yields \$14. How many of each product should be loaded into the truck for maximum profit?



Before Solving

- The stages of the problem are the products. It is assumed that the first, then the second, and finally the third product will be loaded.
- Deciding how much of each product to load constitutes a stage of the problem.
- Decisions are how many units to load for the **product handled at that stage**.
- States represent the remaining empty space in the truck.



Solution

- We solve our problem using the backward recursion method.
- Since we start from the end, we first perform the operations for the 3rd product. We assume that the first and second products have already been loaded, and the third product will be loaded into the remaining space in the truck.
- We do not know how much of the first two products have been loaded. The truck could be completely empty, partially full, or completely filled.



Solution

$$V_3^*(0) = 0 + 0 = 0$$

$$V_3^*(1) = 14 + 0 = 14$$

$$V_3^*(2) = 28 + 0 = 28$$

$$V_3^*(3) = 42 + 0 = 42$$

$$V_3^*(4) = 56 + 0 = 56$$



Solution

$$V_2^*(0) = IC_0^0 + V_3^*(0) = 0 + 0 = 0$$

$$V_2^*(1) = IC_1^0 + V_3^*(1) = 0 + 14 = 14$$

$$V_2^*(2) = IC_2^0 + V_3^*(2) = 0 + 28 = 28$$



Solution

$$V_2^*(3) = \max \left\{ \begin{array}{l} IC_3^0 + V_3^*(3) = 0 + 42 = 42 \\ IC_3^1 + V_3^*(0) = 47 + 0 = 47^* \end{array} \right\} = 47$$
$$V_2^*(4) = \max \left\{ \begin{array}{l} IC_4^0 + V_3^*(4) = 0 + 56 = 56 \\ IC_4^1 + V_3^*(1) = 47 + 14 = 61^* \end{array} \right\} = 61$$



Solution

$$V_1^*(4) = \max \left\{ \begin{array}{lll} IC_4^0 + V_2^*(4) & = 0 + 61 & = 61 \\ IC_4^1 + V_2^*(2) & = 31 + 28 & = 59 \\ IC_4^2 + V_2^*(0) & = 62 + 0 & = 62^* \end{array} \right\} = 62$$



Solution

- The solution to the problem is to load 2 units of the first product.



Homework

A person preparing for a hike can pack food, a first aid kit, and clothing into their backpack. The capacity of the backpack is 300 cm^3 . Each unit of food takes up 100 cm^3 , each first aid kit takes up 25 cm^3 , and each unit of clothing takes up 50 cm^3 . Based on experience, the hiker assigns importance scores of 3, 4, and 5 to food, first aid kits, and clothing, respectively. Furthermore, they know that at least one of each item must be taken, and there is no need to take more than two first aid kits. Which items should be packed to maximize the total importance score in the backpack?



Workforce Planning Problems

- Workforce planning is required in project-based settings or in situations where future labor demand can be forecasted in advance.
- Decisions may include the number of workers to hire or lay off in each period, as well as scheduling their shifts.
- Decision-makers aim to determine a workforce plan that minimizes total labor-related costs.



Objective Function Components

- Hiring costs include expenses such as training, onboarding, and administrative processing.
- Worker compensation may consist of base wages and bonuses.
- Severance payments represent the cost of layoffs.
- In addition, opportunity costs may arise when workers are idle.



An Example

A construction company requires 5, 7, 8, 4, and 6 workers over the next five weeks, respectively. The company pays \$200 per week to each worker employed on-site. Workers who are retained but not assigned to work incur a weekly cost of \$300 (including opportunity costs).

Hiring in any given week incurs a fixed administrative and training cost of \$400, regardless of the number of workers hired.

Assuming that the company currently has no workers, determine the optimal workforce plan over the five-week horizon.



Before Solving

- The stages of the problem are the weeks (decision periods).
- Decisions correspond to the number of workers to hire in each week.
- The state is the number of workers available at the beginning of each week.



Solution Approach

- We solve the problem using backward recursion.
- Since we start from the end, we first determine how many workers should be employed in Week 5, assuming that the first four weeks have already passed.
- For each possible state, we evaluate all feasible decisions.
- We do not know exactly how many workers are carried over from the previous week. However, since the demand in Week 4 is 4 workers, at least 4 workers must have been employed.
- Since the maximum workforce requirement in earlier periods is 8, we can assume that no more than 8 workers are carried over.



Solution Approach

- When determining possible states in subsequent periods, we consider not only past workforce requirements but also future demand.
- The minimum number of workers carried over must be at least equal to the number of workers employed in the previous period.
- However, the maximum number of workers carried over may exceed past workforce levels, since hiring in advance can help avoid high future hiring costs.



Solution

$$V_5^*(4) = IC_4^2 + V_6^*(6) = 400 + 200 * 6 + 0 = 1600$$

$$V_5^*(5) = IC_5^1 + V_6^*(6) = 400 + 200 * 6 + 0 = 1600$$

$$V_5^*(6) = IC_6^0 + V_6^*(6) = 200 * 6 + 0 = 1200$$

$$V_5^*(7) = IC_7^{-1} + V_6^*(6) = 200 * 6 + 0 = 1200$$

$$V_5^*(8) = IC_8^{-2} + V_6^*(6) = 200 * 6 + 0 = 1200$$



Solution

$$V_4^*(8) = \min \left\{ \begin{array}{l} IC_8^{-4} + V_5^*(4) = 4 * 200 + 1600 \\ \quad \quad \quad = 2400^* \\ IC_8^{-2} + V_5^*(6) = 4 * 200 + 2 * 300 + 1200 \\ \quad \quad \quad = 2600 \end{array} \right\}$$

$$= 2400$$



Solution

$$V_3^*(7) = IC_7^1 + V_4^*(8) = 400 + 8 * 200 + 2400 = 4400$$

$$V_3^*(8) = IC_8^0 + V_4^*(8) = 8 * 200 + 2400 = 4000$$



Solution

$$V_2^*(5) = \min \left\{ \begin{array}{l} IC_5^2 + V_3^*(7) = 400 + 7 * 200 + 4400 \\ = 6200 \\ IC_5^3 + V_3^*(8) = 400 + 7 * 200 + 300 + 4000 \\ = 6100^* \end{array} \right\}$$

$$= 6100$$



Solution

$$V_2^*(7) = \min \left\{ \begin{array}{l} IC_7^0 + V_3^*(7) = 7 * 200 + 4400 \\ \quad \quad \quad = 5800^* \\ IC_7^1 + V_3^*(8) = 400 + 7 * 200 + 300 + 4000 \\ \quad \quad \quad = 6100 \end{array} \right\}$$

$$= 5800$$



Solution

$$V_2^*(8) = \min \left\{ \begin{array}{l} IC_8^{-1} + V_3^*(7) = 7 * 200 + 4400 \\ \quad \quad \quad = 5800 \\ IC_8^0 + V_3^*(8) = 7 * 200 + 300 + 4000 \\ \quad \quad \quad = 5700^* \end{array} \right\}$$

$$= 5700$$



Solution

$$V_1^*(0) = \min \left\{ \begin{array}{l} IC_0^5 + V_2^*(5) = 400 + 5 * 200 + 6100 \\ \quad \quad \quad = 7500^* \\ IC_0^7 + V_2^*(7) = 400 + 5 * 200 + 2 * 300 + 5800 \\ \quad \quad \quad = 7800 \\ IC_0^8 + V_2^*(8) = 400 + 5 * 200 + 3 * 300 + 5700 \\ \quad \quad \quad = 8000 \end{array} \right\}$$

$$= 7500$$



Optimal Policy

- The optimal solution is:
 - Hire 5 workers in Week 1
 - Hire 3 workers in Week 2
 - No hiring or layoffs in Week 3
 - Lay off 4 workers in Week 4
 - Hire 2 workers in Week 5



Homework

A tourism company organizes one-week trips to Egypt. Rental vehicles are used for group tours. Over the next four weeks, the company will require 7, 4, 7, and 8 vehicles, respectively.

Signing a rental contract incurs a fixed administrative cost of \$500, regardless of the number of vehicles or the duration. In addition, the weekly rental cost per vehicle is \$220.

Determine how many vehicles should be rented or returned each week.



Equipment / Asset Replacement Problems

- As machines age, their productivity decreases and maintenance costs increase.
- However, replacing all equipment every year may lead to prohibitively high acquisition costs.
- Decision-makers aim to determine the replacement policy that maximizes total profit.



Objective Function Components

- Revenues generated by machines depend on their age.
- Maintenance and operating costs also vary with age.
- In addition, the salvage value of the machine must be considered, since old machines can be sold in the second-hand market or as scrap.
- Finally, the purchase cost of a new machine must be included when replacement occurs.



An Example

A company owns a 3-year-old machine and wants to determine in which years it should be replaced over the next 4 years. The company knows that a 6-year-old machine must be replaced.

A new machine can be purchased for \$100,000. The revenues, operating and maintenance costs, and salvage values depending on the machine's age are given in the following table.



Data

Age	Revenue	Operating Cost	Salvage Value
0	20,000	200	—
1	19,000	600	80,000
2	18,500	1,200	60,000
3	17,200	1,500	50,000
4	15,500	1,700	30,000
5	14,000	1,800	10,000
6	-	-	5,000



Before Solving

- The stages of the problem are years (decision periods).
- The decision in each stage is whether to replace the machine or keep it.
- The state is the age of the machine at the beginning of each year.



Solution Approach

- We solve the problem using backward recursion.
- Starting from the last stage, we evaluate whether to replace or keep the machine for each possible age in Year 4.
- For each possible state, all feasible decisions are evaluated.



Solution Approach

- Since the machine is currently 3 years old, its age in Year 4 can be at most 6.
- Similarly, in Year 3 the maximum age is 5, and in Year 2 the maximum age is 4.
- Decisions are denoted as Y (replace) and K (keep).
- Unlike previous problems, the salvage value in the final period implies that the future value term $V_{t+1}(s')$ must still be considered even in the last stage.



Solution

$$V_4^*(6) = IC_6^Y + V_5^*(1) = (20,000 - 200 - 100,000 + 5,000) + 80,000 \\ = 4,800^*$$

$$V_4^*(5) = \max \left\{ \begin{array}{ll} IC_5^Y + V_5^*(1) & = (20,000 - 200 - 100,000 + 10,000) + 80,000 \\ & = 9,800 \\ IC_5^K + V_5^*(6) & = (14,000 - 1,800) + 5,000 \\ & = 17,200^* \end{array} \right\} \\ = 17,200$$



Solution

$$V_4^*(4) = \max \left\{ \begin{array}{ll} IC_4^Y + V_5^*(1) & = (20,000 - 200 - 100,000 + 30,000) + 80,000 \\ & = 29,800^* \\ IC_4^K + V_5^*(5) & = (15,500 - 1,700) + 10,000 \\ & = 23,800 \end{array} \right\}$$

$$= 29,800$$

$$V_4^*(3) = \max \left\{ \begin{array}{ll} IC_3^Y + V_5^*(1) & = (20,000 - 200 - 100,000 + 50,000) + 80,000 \\ & = 49,800^* \\ IC_3^K + V_5^*(4) & = (17,200 - 1,500) + 30,000 \\ & = 45,700 \end{array} \right\}$$

$$= 49,800$$



Solution

$$V_4^*(2) = \max \left\{ \begin{array}{l} IC_2^Y + V_5^*(1) = (20,000 - 200 - 100,000 + 60,000) + 80,000 \\ \quad = 59,800 \\ IC_2^K + V_5^*(3) = (18,500 - 1,200) + 50,000 \\ \quad = 67,300^* \end{array} \right\}$$

$$= 67,300$$

$$V_4^*(1) = \max \left\{ \begin{array}{l} IC_1^Y + V_5^*(1) = (20,000 - 200 - 100,000 + 80,000) + 80,000 \\ \quad = 79,800^* \\ IC_1^K + V_5^*(2) = (19,000 - 600) + 60,000 \\ \quad = 78,400 \end{array} \right\}$$

$$= 79,800$$



Solution

$$V_3^*(5) = \max \left\{ \begin{array}{ll} IC_5^Y + V_4^*(1) & = (20,000 - 200 - 100,000 + 10,000) + 79,800 \\ & = 9,600 \\ IC_5^K + V_4^*(6) & = (14,000 - 1,800) + 4,800 \\ & = 17,000^* \end{array} \right\}$$

$$= 17,000$$

$$V_3^*(4) = \max \left\{ \begin{array}{ll} IC_4^Y + V_4^*(1) & = (20,000 - 200 - 100,000 + 30,000) + 79,800 \\ & = 29,600 \\ IC_4^K + V_4^*(5) & = (15,500 - 1,700) + 17,200 \\ & = 31,000^* \end{array} \right\}$$

$$= 31,000$$



Solution

$$V_3^*(3) = \max \left\{ \begin{array}{ll} IC_3^Y + V_4^*(1) & = (20,000 - 200 - 100,000 + 50,000) + 79,800 \\ & = 49,600^* \\ IC_3^K + V_4^*(4) & = (17,200 - 1,500) + 29,800 \\ & = 45,500 \end{array} \right\}$$

$$= 49,600$$

$$V_3^*(2) = \max \left\{ \begin{array}{ll} IC_2^Y + V_4^*(1) & = (20,000 - 200 - 100,000 + 60,000) + 79,800 \\ & = 59,600 \\ IC_2^K + V_4^*(3) & = (18,500 - 1,200) + 49,800 \\ & = 67,100^* \end{array} \right\}$$

$$= 67,100$$



Solution

$$V_3^*(1) = \max \left\{ \begin{array}{ll} IC_1^Y + V_4^*(1) & = (20,000 - 200 - 100,000 + 80,000) + 79,800 \\ & = 79,600 \\ IC_1^K + V_4^*(2) & = (19,000 - 600) + 67,300 \\ & = 85,700^* \end{array} \right\}$$

$$= 85,700$$



Solution

$$V_2^*(4) = \max \left\{ \begin{array}{l} IC_4^Y + V_3^*(1) = (20,000 - 200 - 100,000 + 30,000) + 85,700 \\ \quad = 35,500^* \\ IC_4^K + V_3^*(5) = (15,500 - 1,700) + 17,000 \\ \quad = 30,800 \end{array} \right\}$$

$$= 35,500$$

$$V_2^*(3) = \max \left\{ \begin{array}{l} IC_3^Y + V_3^*(1) = (20,000 - 200 - 100,000 + 50,000) + 85,700 \\ \quad = 55,500^* \\ IC_3^K + V_3^*(4) = (17,200 - 1,500) + 31,000 \\ \quad = 46,700 \end{array} \right\}$$

$$= 55,500$$



Solution

$$V_2^*(2) = \max \left\{ \begin{array}{ll} IC_2^Y + V_3^*(1) & = (20,000 - 200 - 100,000 + 60,000) + 85,700 \\ & = 65,500 \\ IC_2^K + V_3^*(3) & = (18,500 - 1,200) + 49,600 \\ & = 66,900^* \end{array} \right\}$$

$$= 66,900$$

$$V_2^*(1) = \max \left\{ \begin{array}{ll} IC_1^Y + V_3^*(1) & = (20,000 - 200 - 100,000 + 80,000) + 85,700 \\ & = 85,500^* \\ IC_1^K + V_3^*(2) & = (19,000 - 600) + 67,100 \\ & = 85,500^* \end{array} \right\}$$

$$= 85,500$$



Solution

$$V_1^*(3) = \max \left\{ \begin{array}{l} IC_3^Y + V_2^*(1) = (20,000 - 200 - 100,000 + 50,000) + 85,500 \\ \quad \quad \quad = 55,300^* \\ IC_3^K + V_2^*(4) = (17,200 - 1,500) + 35,500 \\ \quad \quad \quad = 51,200 \end{array} \right\}$$

$$= 55,300$$



Optimal Policy

- The optimal solution starts with replacing the machine in Year 1.
- In Year 2, both replacing and keeping the machine are optimal decisions.
- If the machine is replaced in Year 2, then no replacement should occur in Years 3 and 4.
- If the machine is not replaced in Year 2, then it should not be replaced in Year 3 but must be replaced in Year 4.
- The optimal total profit is \$55,300 in both cases.



Remark

- Note: In Year 2, the machine can only be either 1 or 4 years old (similar constraints apply in other years).
- Therefore, it is not necessary to compute the value function for all possible ages.
- However, all cases are shown here for clarity and instructional purposes.



Homework

A company owns a 2-year-old machine and wants to determine in which years it should be replaced over the next 5 years. The company knows that a 4-year-old machine must be replaced.

A new machine can be purchased for \$200,000. The revenues, operating costs, and salvage values depending on the machine's age are given below:

Age	Revenue	Operating Cost	Salvage Value
0	40,000	1,000	—
1	35,000	2,000	150,000
2	30,000	4,000	120,000
3	25,000	6,000	80,000
4	-	-	20,000



Inventory Problems

- Holding inventory is essential for many businesses.
- In addition to operational necessity, inventory often constitutes one of the most significant cost components, which increases the importance of inventory management.
- In basic inventory problems, two key decisions are made: when to order and how much to order.



Inventory Costs

- In this course, we focus on two main cost components.
- The first is the **holding cost**, which is incurred for each unit kept in inventory for each period.
- For example, if the unit holding cost is 2 TL per product per period, then holding 10 units for 3 periods results in a total cost of $2 \times 3 \times 10 = 60$ TL.



Inventory Costs

- The second cost component is the **fixed ordering (setup) cost**.
- This cost is incurred whenever an order is placed or a production batch is initiated.
- It includes transportation, administrative processing, and machine setup costs such as preparation, warm-up, and cleaning.
- It is independent of the order or production quantity. Producing or ordering 1 unit or 1,000 units does not change the fixed cost.



An Example

A firm forecasts its demand over the next five periods as 20, 10, 30, 60, and 10 thousand units, respectively. The unit holding cost is 3 TL, and the fixed ordering cost is 80 thousand TL.

Production can only be carried out in multiples of 20 thousand units, and the maximum production capacity per period is 60 thousand units. The warehouse capacity is 30 thousand units.

Assuming the firm currently has an initial inventory of 10 thousand units, determine the optimal production and inventory plan over the five periods.



Before Solving

- The stages of the problem are the periods (decision epochs).
- Decisions correspond to how much to produce in each period.
- The state is the inventory level at the beginning of each period.



Solution Approach

- We solve the problem using backward recursion.
- Starting from the last period, we first determine the production decision for Period 5, assuming earlier periods have already occurred.
- Since the inventory carried over from the previous period is unknown, we evaluate all possible states and feasible decisions.
- The warehouse capacity implies that no more than 30 thousand units can be carried over.
- Note: Costs below are expressed in thousands of TL.



Solution

$$V_5^*(0) = IC_0^{20} + V_6^*(10) = (80 + 10 * 3) + 0 = 110$$

$$V_5^*(10) = IC_{10}^0 + V_6^*(0) = (0 + 0) + 0 = 0$$

$$V_5^*(20) = IC_{20}^0 + V_6^*(10) = (0 + 10 * 3) + 0 = 30$$

$$V_5^*(30) = IC_{30}^0 + V_6^*(20) = (0 + 20 * 3) + 0 = 60$$



Solution

$$V_4^*(0) = IC_0^{60} + V_5^*(0) = (80 + 0) + 110 = 190$$

$$V_4^*(10) = IC_{10}^{60} + V_5^*(10) = (80 + 10 * 3) + 0 = 110$$

$$V_4^*(20) = \min \left\{ \begin{array}{ll} IC_{20}^{40} + V_5^*(0) & = (80 + 0) + 110 = 190 \\ IC_{20}^{60} + V_5^*(20) & = (80 + 20 * 3) + 30 = 170^* \end{array} \right\}$$

$$= 170$$

$$V_4^*(30) = \min \left\{ \begin{array}{ll} IC_{30}^{40} + V_5^*(10) & = (80 + 10 * 3) + 0 = 110^* \\ IC_{30}^{60} + V_5^*(30) & = (80 + 30 * 3) + 60 = 230 \end{array} \right\}$$

$$= 110$$



Solution

$$V_3^*(0) = \min \left\{ \begin{array}{l} IC_0^{40} + V_4^*(10) = (80 + 10 * 3) + 110 = 220^* \\ IC_0^{60} + V_4^*(30) = (80 + 30 * 3) + 110 = 280 \end{array} \right\}$$

$$= 220$$

$$V_3^*(10) = \min \left\{ \begin{array}{l} IC_{10}^{20} + V_4^*(0) = (80 + 0) + 190 = 270^* \\ IC_{10}^{40} + V_4^*(20) = (80 + 20 * 3) + 170 = 310 \\ IC_{10}^{60} + V_4^*(30) = (80 + 40 * 3) + 110 = 310 \end{array} \right\}$$

$$= 270$$



Solution

$$V_3^*(20) = \min \left\{ \begin{array}{ll} IC_{20}^{20} + V_4^*(10) & = (80 + 10 * 3) + 110 = 220^* \\ IC_{20}^{40} + V_4^*(30) & = (80 + 30 * 3) + 110 = 280 \\ IC_{20}^{60} + V_4^*(30) & = (80 + 50 * 3) + 110 = 340 \end{array} \right\}$$

$$= 220$$

$$V_3^*(30) = \min \left\{ \begin{array}{ll} IC_{30}^0 + V_4^*(0) & = (0 + 0) + 190 = 190^* \\ IC_{30}^{20} + V_4^*(20) & = (80 + 20 * 3) + 170 = 310 \\ IC_{30}^{40} + V_4^*(30) & = (80 + 40 * 3) + 110 = 310 \\ IC_{30}^{60} + V_4^*(30) & = (80 + 60 * 3) + 110 = 370 \end{array} \right\}$$

$$= 190$$



Solution

$$V_2^*(0) = \min \left\{ \begin{array}{ll} IC_0^{20} + V_3^*(10) & = (80 + 10 * 3) + 270 = 380 \\ IC_0^{40} + V_3^*(30) & = (80 + 30 * 3) + 190 = 360^* \\ IC_0^{60} + V_3^*(30) & = (80 + 50 * 3) + 190 = 420 \end{array} \right\}$$

$$= 360$$

$$V_2^*(10) = \min \left\{ \begin{array}{ll} IC_{10}^0 + V_3^*(0) & = (0 + 0) + 220 = 220^* \\ IC_{10}^{20} + V_3^*(20) & = (80 + 20 * 3) + 220 = 360 \\ IC_{10}^{40} + V_3^*(30) & = (80 + 40 * 3) + 190 = 390 \\ IC_{10}^{60} + V_3^*(30) & = (80 + 60 * 3) + 190 = 450 \end{array} \right\}$$

$$= 220$$



Solution

$$V_2^*(20) = \min \left\{ \begin{array}{ll} IC_{20}^0 + V_3^*(10) & = (0 + 10 * 3) + 270 = 300^* \\ IC_{20}^{20} + V_3^*(30) & = (80 + 30 * 3) + 190 = 360 \\ IC_{20}^{40} + V_3^*(30) & = (80 + 50 * 3) + 190 = 420 \\ IC_{20}^{60} + V_3^*(30) & = (80 + 70 * 3) + 190 = 480 \end{array} \right\}$$

$$= 300$$

$$V_2^*(30) = \min \left\{ \begin{array}{ll} IC_{30}^0 + V_3^*(20) & = (0 + 20 * 3) + 220 = 280^* \\ IC_{30}^{20} + V_3^*(30) & = (80 + 40 * 3) + 190 = 390 \\ IC_{30}^{40} + V_3^*(30) & = (80 + 60 * 3) + 190 = 450 \\ IC_{30}^{60} + V_3^*(30) & = (80 + 80 * 3) + 190 = 510 \end{array} \right\}$$

$$= 280$$



Solution

$$V_1^*(10) = \min \left\{ \begin{array}{l} IC_{10}^{20} + V_2^*(10) = (80 + 10 * 3) + 220 = 390^* \\ IC_{10}^{40} + V_2^*(30) = (80 + 30 * 3) + 280 = 450 \\ IC_{10}^{60} + V_2^*(30) = (80 + 50 * 3) + 340 = 570 \end{array} \right\}$$
$$= 390$$



Optimal Policy

- The optimal solution is to produce 20, 0, 40, 60, and 0 thousand units over the five periods, respectively.
- Therefore, production occurs only in Periods 1, 3, and 4.



Homework

A firm forecasts its demand over the next five periods as 100, 200, 500, 300, and 500 units, respectively. The unit holding cost is 2 TL, and the fixed ordering cost is 1000 TL.

Production is carried out in batches, where each batch consists of 200 units. The firm can produce at most three batches in a single period. Determine the optimal production plan using dynamic programming.



Homework

The product produced by the firm has a constant weekly demand of 500 units. Demand is satisfied through orders placed with a supplier in multiples of 1000 units.

In case of a stockout, up to 500 units of unmet demand can be backordered and satisfied at the beginning of the next week at a cost of 8 TL per unit.

The unit holding cost is 1 TL, and the fixed ordering cost is 5000 TL. Determine the optimal ordering plan over a one-year horizon.



Homework

A firm forecasts its demand over the next five months as 100, 200, 500, 300, and 500 units, respectively. The unit holding cost is 2 TL, and the fixed ordering cost is 1000 TL.

Production is carried out in batches. **The production lead time is one month, meaning that orders placed in the current month become available in the next month.**

Each batch consists of 200 units, and the firm can produce at most three batches in a single period.

If the initial inventory is 200 units, determine the optimal production plan using dynamic programming.



Homework

A firm sells two different products and forecasts their demands over the next five months as $(20, 10, 30, 60, 10)$ and $(20, 30, 30, 20, 30)$, respectively.

The unit holding cost is 3 TL, and the fixed ordering cost is 80 TL. Production is carried out in batches of 20 units, and at most three batches can be produced in a single period.

The end-of-period storage capacities for the two products are 30 and 20 units, respectively.

If the initial inventory for both products is 10 units, determine the optimal production plan for each period using dynamic programming.



An Example

A construction company requires 5, 7, 11, 6, 9, 3, 4, 11, and 8 workers over nine weeks, respectively.

The company pays 200 dollars per week to each worker employed on-site. Workers who are retained but not actively working incur a cost of 300 dollars per week due to opportunity costs.

Hiring in any week incurs a fixed cost of 400 dollars, independent of the number of workers hired. Firing costs 500 dollars per worker.

Assuming the company currently has no workers, determine the optimal workforce plan over the planning horizon.



Homework

A distribution company needs to deliver low-volume products to 8 customers. A single vehicle must visit all customers and return to the depot.

Determine the shortest route based on the distance matrix below using an optimization approach.

	Depot	C1	C2	C3	C4	C5	C6	C7	C8
Depot	9999	11	13	14	14	12	15	23	17
C1	12	9999	3	4	4	12	12	15	11
C2	14	2	9999	2	4	10	10	13	9
C3	15	4	3	9999	2	8	8	11	7
C4	15	4	3	2	9999	10	10	13	9
C5	10	12	11	8	10	9999	4	15	11
C6	11	10	9	7	8	2	9999	13	9
C7	24	16	15	12	14	15	16	9999	6
C8	21	11	10	8	9	11	11	6	9999



Homework

A firm has observed that different employees generate different levels of customer satisfaction when interacting with customers.

Using the benefit matrix below, assign customers to employees such that each employee is assigned at most two customers.

	C1	C2	C3	C4	C5	C6	C7	C8
E1	94	82	99	70	50	72	50	94
E2	94	93	65	75	79	99	77	84
E3	66	59	64	95	89	75	87	63
E4	50	53	100	64	70	79	61	65
E5	75	89	72	88	94	60	55	71



İçindekiler

1 Markov Decision Processes

- Definition
- Application Areas
- Terminology
- Markov Decision Process Models

2 What is Dynamic Programming?

- A Simple Definition
- The Origins of Dynamic Programming
- Recursive Dynamic Programming Structure

3 DP with Backward and Forward Recursion

- Hamilton-Jacobi-Bellman

Equation

4 Deterministic DP Examples

- Cargo Loading Problems
- Workforce Planning Problems
- Equipment Replacement Problems
- Inventory Problems
- Practice Problems

5 Stochastic Markov Decision Processes

- Stochastic MDPs
- Solution Approach for Stochastic Problems
- An Example



What Does Stochastic Mean?

- The exact values of all system parameters may not be known with certainty.
- If at least one parameter is uncertain, the problem is no longer deterministic.
- Uncertainty may arise in the form of discrete scenarios with known probabilities; such problems are called scenario-based problems.
- Alternatively, uncertainty may be represented through probability distributions of parameters. Both cases fall under stochastic problems.



What Are Stochastic Processes?

- Stochastic processes are systems that evolve over time in a probabilistic manner.
- The condition of the system at any point in time is called a **state**.
- The system transitions from one state to another with certain probabilities.
- These probabilities are represented by the **transition probability matrix**.



Markov Chains and Transition Probabilities

- A Markov chain $\{X_0, X_1, X_2, \dots\}$ is defined by a transition matrix P .
- P_{ij} denotes the probability of transitioning from state i to state j .
- **Markov Property:** The next state depends only on the current state, not on past states.
- This property is known as the **memoryless property**.



Types of States: Transient vs Recurrent

- A state is **recurrent** if the system will return to it with probability 1.
- A state is **transient** if there is a positive probability of never returning to it.
- In other words, once a transient state is left, the system may never come back.



Periodicity

- The period of a state i is $k \geq 1$ if returns to state i are only possible in multiples of k steps.
- If $k = 1$, the state is **aperiodic**.
- If $k > 1$, the state is **periodic**.
- If all states are aperiodic, the Markov chain is called aperiodic.



Irreducibility

- A Markov chain is **irreducible** if every state can be reached from every other state with positive probability.
- This means no state is isolated from the rest of the system.



Absorbing States

- A state i is called an **absorbing state** if $P_{ii} = 1$.
- Once the system enters this state, it cannot leave.
- Absorbing states are central to the study of absorbing Markov chains.



Types of Recurrent States

- If a state is recurrent, the probability of returning to it is 1.
- Recurrent states are classified as:
 - **Positive Recurrent:** The expected return time is finite.
 - **Null Recurrent:** The expected return time is infinite.



Ergodicity

- A state is **ergodic** if it is both positive recurrent and aperiodic.
- If all states are ergodic, the Markov chain is ergodic.
- In ergodic chains, the system converges to a steady-state (stationary) distribution.



Stationary Distribution

- The stationary distribution describes long-run state probabilities.
- The vector π satisfies:

$$\pi P = \pi, \quad \sum_i \pi_i = 1.$$

- In ergodic chains, this distribution is unique and the system converges to it.



Summary of State Types

- **Transient:** Possible to never return.
- **Positive Recurrent:** Guaranteed return with finite expected time.
- **Null Recurrent:** Guaranteed return but infinite expected time.
- **Ergodic:** Positive recurrent and aperiodic, admits a stationary distribution.



Additional Remarks

- Markov chains form the foundation of stochastic decision processes.
- Uncertainty in stochastic systems is modeled probabilistically.
- Positive recurrent chains exhibit stable long-term behavior.
- Transient states typically occur in early stages before convergence.
- Ergodic chains allow prediction of long-run average behavior.



Expected Value

- Consider a gambling game.
- The expected value is the long-run average outcome per trial if the experiment is repeated infinitely many times.
- $\sum(p_i \cdot r_i)$



Bellman Equation for Stochastic Problems

$$\begin{aligned} V_t^*(s) &\leftarrow \min_a \mathbb{E}\{IC_s^a + V_{t+1}(s')\} \\ &\leftarrow \min_a \left\{ \mathbb{E}\{IC_s^a\} + \sum_{s'} \beta_{s,s'}^a V_{t+1}^*(s') \right\} \\ &\leftarrow \min_a \left\{ \sum_i p_{s,i}^a r_{s,i}^a + \sum_{s'} \beta_{s,s'}^a V_{t+1}^*(s') \right\} \end{aligned}$$



An Example

A farmer analyzes the soil before each planting season. The soil condition is classified as **good**, **moderate**, or **poor**.

During the season, the soil condition may change due to environmental factors. These transition probabilities and the corresponding revenues are given in matrices P and R .



An Example

$$P = \begin{pmatrix} 0.2 & 0.5 & 0.3 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R = \begin{pmatrix} 7 & 6 & 3 \\ 0 & 5 & 1 \\ 0 & 0 & -1 \end{pmatrix}$$



An Example

The farmer also has the option to fertilize the soil. In this case, the matrices change to P' and R' .



An Example

$$P = \begin{pmatrix} 0.2 & 0.5 & 0.3 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R = \begin{pmatrix} 7 & 6 & 3 \\ 0 & 5 & 1 \\ 0 & 0 & -1 \end{pmatrix}$$

$$P' = \begin{pmatrix} 0.3 & 0.6 & 0.1 \\ 0.1 & 0.6 & 0.3 \\ 0.05 & 0.4 & 0.55 \end{pmatrix}$$

$$R' = \begin{pmatrix} 6 & 5 & -1 \\ 7 & 4 & 0 \\ 6 & 3 & -2 \end{pmatrix}$$



Before Solving

- Stages correspond to periods (decision epochs).
- The decision is whether to fertilize in each period.
- The state is the soil condition at the beginning of each period.



Solution Approach

- We solve the problem using backward recursion.
- Starting from the last period, we determine the optimal decision assuming earlier periods have already occurred.
- Since the incoming state is uncertain, we evaluate all possible states and decisions.



Thanks for listening!

See you in the next lecture...

Dr. Mustafa Çimen

